

BRAC University

Department of Electrical and Electronic Engineering
CSE162/EEE103IL/ECE103IL: Computer Programming Laboratory

Final Project

Marks: 25

Project Title: *EEE Pocket Tool – A Multi-Functional Circuit Assistant in C*

Background and Motivation

Modern engineers often need to perform quick electrical calculations while working in labs, field testing, or during prototyping. To assist with this, you are tasked with developing a menu-driven C program that acts as a toolbox—an **EEE Pocket Tool**—for solving common circuit problems. This tool is designed to be a helpful companion for students and engineers alike.

Objective

Create a user-friendly, console-based C application that allows users to select from a set of tools based on their needs. Each tool will prompt for the required inputs and provide relevant electrical outputs in a structured and readable format. The following demo menu is provided for reference only, and the details of each tool are described later. Since this is an open-ended project, you are encouraged to modify the menu and its functionality according to your preferences, while ensuring that all the tools' functionalities are implemented.

```
===== EEE Pocket Tool =====
1. Series Circuit Solver
2. Parallel Circuit Solver
3. Resistor Value Determiner
4. Equivalent Resistance Calculator
5. Voltage Divider Calculator
0. Exit
Enter your choice:
```

Tool Descriptions and Equations

1. Series Circuit Solver

Description: This tool solves circuits with one voltage source and n resistors in series. The program will prompt the user to input the source voltage V_s , the number of resistors and the values of each resistor.

Inputs:

- Source voltage V_s
- Number of resistors
- Resistor values $R_1, R_2, \dots, R_n$

Outputs:

- Equivalent resistance R_{eq}
- Current flowing in the circuit I
- Voltage drop across each resistor V_i

Equations:

$$R_{\text{eq}} = R_1 + R_2 + \dots + R_n$$

$$I = \frac{V_s}{R_{\text{eq}}}$$

$$V_i = I \cdot R_i \quad (\text{Voltage across each resistor})$$

Hint: Store resistor values in an array and iterate to calculate the total resistance, current and voltage drops.

2. Parallel Circuit Solver

Description: This tool solves circuits with one voltage source and n resistors in parallel. The program will prompt the user to input the source voltage V_s , the number of resistors, and the values of each resistor.

Inputs:

- Source voltage V_s
- Number of resistors
- Resistor values $R_1, R_2, \dots, R_n$

Outputs:

- Equivalent resistance R_{eq}
- Total current I
- Current through each resistor I_i

Equations:

$$\frac{1}{R_{\text{eq}}} = \sum_{i=1}^n \frac{1}{R_i}$$

$$I = \frac{V_s}{R_{\text{eq}}}$$

$$I_i = \frac{V_s}{R_i} \quad (\text{Current through each resistor})$$

Hint: Store resistor values in an array and use a loop to compute the reciprocal sum for the equivalent resistance.

3. Resistor Value Determiner

Description: This tool determines the resistance value of a resistor based on the 4-band color code. The program will prompt for the color of each band, and based on the color code, it will calculate the resistance value.

Inputs:

- Band 1 color (first digit)
- Band 2 color (second digit)
- Multiplier color

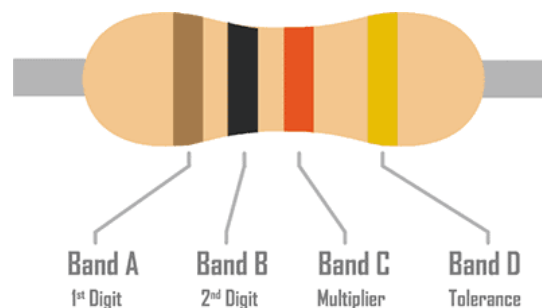
Output:

- Resistance value

Resistor Color Code Table

Color	Digit	Multiplier	Tolerance
Black	0	$\times 1$	—
Brown	1	$\times 10$	$\pm 1\%$
Red	2	$\times 100$	$\pm 2\%$
Orange	3	$\times 1,000$	—
Yellow	4	$\times 10,000$	—
Green	5	$\times 100,000$	$\pm 0.5\%$
Blue	6	$\times 1,000,000$	$\pm 0.25\%$
Violet	7	$\times 10,000,000$	$\pm 0.1\%$
Gray	8	$\times 100,000,000$	$\pm 0.05\%$
White	9	$\times 1,000,000,000$	—
Gold	—	$\times 0.1$	$\pm 5\%$
Silver	—	$\times 0.01$	$\pm 10\%$
None	—	—	$\pm 20\%$

Example Resistor:



The first band is brown (1), the second band is black (0), and the third band is orange ($\times 1000$). So the resistance value = (1st band value $\times 10$ + 2nd band value) $\times$ 3rd band multiplier = $(1 \times 10 + 0) \times 1000 = 10000$.

Hint: Try to make an array storing all the colors. Form the array in such a way that Black stays at index 0, Brown in index 1, and so on. So when the user inputs a band color, finding that color's index will correspond to the value that color denotes. For the multiplier, you can use the condition of 10^{value} . So, if the multiplier band is red, that means value is 2, so $10^2 = 100$, which corresponds to the multiplier value from the table. But for gold and silver, handle those cases separately. You can use 'string.h' library's functions like strcmp().

4. Equivalent Resistance Calculator

Description: This tool calculates the equivalent resistance for a given set of resistors. The user will specify whether the resistors are in series or parallel, input the number of resistors and provide their values.

Inputs:

- Type (series or parallel)
- Number of resistors
- Resistor values

Outputs:

- Equivalent resistance

5. Voltage Divider Calculator

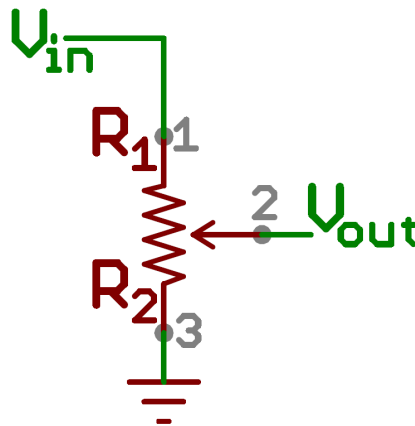
Description: This tool has 2 options, it allows users to either calculate the output voltage for a given input voltage and resistor values or determine the resistor values for a desired output voltage.

- a. Output Voltage Determiner

$$V_{\text{out}} = V_{\text{in}} \cdot \frac{R_2}{R_1 + R_2}$$

Input: V_{in}, R_1, R_2

Output: V_{out}



- b. Resistance Calculator for Desired Output

Given V_{in} and V_{out} , find R_2 :

$$\text{As } R_1 \text{ is known: } R_2 = R_1 \cdot \frac{V_{\text{out}}}{V_{\text{in}} - V_{\text{out}}}$$

Input: $V_{\text{in}}, R_1, V_{\text{out}}$

Output: R_2

Suggestion:

Convert each tool into a function. In the menu-driven code, for a specific input, just call the dedicated function for that tool. To make your code more readable you can follow the following tutorial: <https://www.geeksforgeeks.org/how-to-add-code-block-in-google-docs/> . This is an optional step but will make your presentation better. If you decide to use this tool then use the theme "arduino-light" and make sure "No background" is ticked.

Submission Instructions

- This is a group submission.
- Each group will submit one report only in PDF format.
- The report should be typed in Times New Roman font size 12 and text alignment: Justified.
- For code formatting, you can use any code beautifier tool, such as: <https://codebeautify.org/c-formatter-beautifier>.
- In the report, properly state:
 - The project title
 - The objectives of the project
 - The full code
- Make sure to include comments in your code to explain the thought process behind your work.
- Run all 5 tools at least once and attach screenshots of the outputs.
- Any sort of plagiarism (even partial) will result in 0.